Hacker Holidays 2026 – Day 14
Management Wants a Word

🏝️ TODAY'S ITINERARY

- [ ] Take a closer look at what she left behind
- [ ] Some things aren't as locked away as she thought
- [ ] Find out what she was hiding, and claim the flag

In this challenge from Hacker Holidays, we're asked to do Windows filesystem forensics to retrieve the challenge flag

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```
management-wants-a-word-forensics-hh-day-14
└── KAPE
    └── C
        ├── Users
        │   ├── Default
        │   │   ├── NTUSER.DAT
        │   │   ├── NTUSER.DAT.LOG1
        │   │   └── NTUSER.DAT.LOG2
        │   └── vera
        │       ├── AppData
        │       │   ├── Local
        │       │   │   ├── Google
        │       │   │   │   └── Chrome For Testing
        │       │   │   │       └── User Data
        │       │   │   │           ├── AutofillStates
        │       │   │   │           ├── BrowserMetrics
        │       │   │   │           │   └── BrowserMetrics-6A5D5671-3C8.pma
```

When we unzip the task files for this challenge, we see that it contains a Windows file system dump

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@0xMia · posted 40 min after room unlock

> "ok so apparently a browser will remember things for you that you never told anyone else 💀 not every hidden file needs a password cracker, some of them just need a really good memory also why did Patch tell me this version number 1.26.29 idk what it means :(#HackerHolidays"

From the message included with the challenge description, it's implied that we're supposed to extract secrets from the web browser

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Required Files

- “Login Data” file
- “Local State” file
- “SAM” file
- “SYSTEM” file

To extract secrets from the Google Chrome web browser, we will require a number of different files

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Required Files

- “Login Data” file
- “Local State” file
- “SAM” file
- “SYSTEM” file

The Login Data file is a database file that Chrome uses, and contains the secrets we're looking for

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Required Files

- “Login Data” file
- “Local State” file
- “SAM” file
- “SYSTEM” file

The Local State file is a JSON format file that Chrome uses for user settings and encryption keys

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Required Files

- “Login Data” file
- “Local State” file
- “SAM” file
- “SYSTEM” file

The SAM file is a Windows system file which contains local account usernames and passwords

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Required Files

- “Login Data” file
- “Local State” file
- “SAM” file
- “SYSTEM” file

The SYSTEM file is a Windows system file which contains encryption keys for the SAM file

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```
    "encrypted_key": "RFBBUEkBAAAA0Iyd3wEV0RGMegDAT8KX6wEAAADvGQfJmFtOR7k0E21ganAq
EAAAADQAAABHAG8AbwBnAGwAZQAgAEMAaAByAG8AbQBlACAAZgBvAHIAIABUAGUAcwB0AGkAbgBnAAAAEG
YAAAABAAAgAAAA4t9N2ZWJ6/3gYrwIs9GRKJIs/cW8DXo55B2nY8jabSQAAAAADoAAAAACAAAgAAAAQXy4
66r2xSWddI+G09UlfQvFHsjD1ctlZnvVCL10R9IwAAAArDHeduQIrK4XODPWLS/xsuAyZRpOTbd87RH3lk
p96YIpuSV/fCTMAr5itJphn/BnQAAAAKI1dsBXRgJu8ENjGjStvxSEyReIxqJOfXkKQNoMu7rv/JQfjXhJ
YlCWlr0KDh+1s9zhrgJM8A74VyeZqhD8yXU="
```


Step 1: Extract the Encryption Key from the Local State file. We Grep for a value named “encrypted key”

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```
cat encrypted_key | base64 -d > browser_blob.bin

theshyhat㉿hackerfrogs)-[/tmp/hf/hh/day14]
tail -c +6 browser_blob.bin > file.tmp && mv file.tmp browser_blob.bin
```

Step 2: Convert the encrypted key into a binary file, then modify it for use with the Impacket tool

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```
Administrator:500:aad3b435b51404eeaad3b435b51404ee:1241186a4aac4f34f4bf7ace71b396a
8:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089
c0:::
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:1961c38510b8e33fcdb1879616
d12dfc:::
vera:1000:aad3b435b51404eeaad3b435b51404ee:1241186a4aac4f34f4bf7ace71b396a8:::
[*] Cleaning up...
```

Step 3: Extract the user account password hashes using the Impacket tool

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```
Dictionary cache hit:
* Filename..: /usr/share/wordlists/rockyou.txt
* Passwords.: 14344385
* Bytes.....: 139921507
* Keyspace..: 14344385

1241186a4aac4f34f4bf7ace71b396a8:minivera
```

Step 4: Crack the hash for the Vera user and obtain the password. We'll use the Hashcat tool

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C/Users/<USER_NAME>/AppData/Roaming/Microsoft/Protect/<USER_SID>/<USER_GUID>

To perform the next step, we'll need access to the Windows DPAPI master key file for the Vera user

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```
Decrypted key with User Key (SHA1)
Decrypted key: 0x5e5715ec9b6df5a86e97902692a66d28e691f05d5bc1e0
c07e5004a0179d3a96df2468885a28175b0b02cc064445f116a752d2b3e9d40
```

Step 5: extract the DPAPI key for Vera's account using the Impacket tool. We'll need Vera's SID value as well as their password hash for this, and provide their password

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```
Successfully decrypted data
 0000   20 6A 39 A0 97 13 27 EA  94 87 E4 AE A9 84 4F 5D
 0010   36 70 16 24 56 98 22 76  93 9A 71 26 46 DA 0B 02
```

Step 6: use the decrypted DPAPI and the browser blob to extract the AES key for the Chrome database file. Once again, we use Impacket

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```
URL / Service                    | Username
_______________________________________________________
http://bytelotus.thm:8080/login | VeraSecretVault
```

```
| Password
________________
| Wh4t1sV3raD0inG0nTh1sH0st
```

Step 7: use a script to interact with the Login Data file and extract its secrets

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```
└─$ ls -la backup
-rw-rw-r-- 1 theshyhat theshyhat 104857600
```

Now we have a password, but what do we use this with? There's a backup file located in the Vera user's Documents directory

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```
└─$ ls -la backup
-rw-rw-r-- 1 theshyhat theshyhat 104857600
```

```
└─$ ent backup
Entropy = 7.999998 bits per byte.
```

This backup file is a VeraCrypt file, and there are two indicators that this is the case. The first indicator is the file size, which is evenly divisible by 512. The second indicator is a high entropy score, which is the randomness of the bytes

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```
└─$ sudo cryptsetup tcryptOpen backup backup
Enter passphrase for backup:

┌──(myenv)─(theshyhat㉿hackerfrogs)-[/tmp/hf/
└─$ find /dev -name backup
/dev/mapper/backup
/dev/disk/by-loop-ref/backup
```

We can mount the container using cryptsetup, then locate the mount in the /dev directory

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```
total 8
drwxr-xr-x 5 root root 1024 Dec 31  1969  .
drwxr-xr-x 4 root root 4096 Sep  2 15:46  ..
drwxr-xr-x 2 root root 1024 Aug  4 00:16 '$RECYCLE.BIN'
drwxr-xr-x 2 root root 1024 Aug  4 00:16  secret_financial_documents
drwxr-xr-x 2 root root 1024 Aug  4 00:17 'System Volume Information'
```

Lastly, we mount the drive, then find the flag in the secret_financial_documents directory